Abstract

People love their pets and contrariwise, but there are times you wish to depart your pets reception for long durations alone and this is often a problematic issue. Well, we here propose to a pet daycare robot which will monitor still as feed the dogs or cats in an exceedingly timely manner. Our project is automatic pet monitoring and feeding system using Internet of Things. the stress on choosing this because the title is because, initially to relinquish solution to an issue faced by many of us who keeps the pets. Human interference on the a part of taking care of pet once they are busy is difficult. And hence our system are efficient enough to beat the hurdles faced by humans in taking care of pets. This pet care system could be a complete equipment for monitoring all the pet activities (especially for dog) and also by making the pet be at liberty. Furthermore, the project is subdivided into several modules each of which has their unique feature. The system can feed the pets, monitor the movement of pet and update the status to the owner. Feeding pets responsibly and smartly is difficult for plenty of individuals.The matter becomes especially obvious when the owners have a heavily occupied personal life. When owners don't have time to feed them on time, they will leave the feeder full before leaving.

Contents

List of Abbreviations

1) Wi-Fi - Wireless Fidelity

2) IoT - Internet of things

3) ESP- Espresif

4) FTDI- Future Technology Devices International Limited

5) RX- Reciever

6) TX- Transmitter

7) GND- Ground

8) VCC- Voltage Collector Collector

9) IDE- Integrated Development Environment

10) IEEE- Institute of Electrical and Electronics Engineers

11) PO- Programme Outcomes

12) PSO- Programme Specific Outcomes

13) CO- Course Outcomes

List of Figures

Chapter 1

Introduction

1.1 Overview

Pets are a great blessing in anyone's life and they require special care. The goal of this work is to introduce, design and implement a smart pet system. There are times you need to leave your pets at home alone for long duration and this is an issue. We are proposing to design a daycare robot that can feed and monitor the pet with the help of our smartphone. The Robot is an IoT based robot that is capable of following the path the pet takes ensuring constant monitoring. The robot is integrated with a camera that gives the user live stream footage on their smartphone, They also have a food dispensing system that provides food the pet of varying portions chosen by the user with the help of a slider provided in the Blynk app.

1.2 Objectives

- **Monitoring**: The Robot is an IoT Based robot that is capable of taking care you your pets alone at home. The robot is integrated with a camera that allows for live streaming over IoT platform to get on demand footage of home
- **Live telecast**: We can monitor the pet's actions during the absence of their owner through a camera mounted on a robot which follows the pet when commanded and also give commands.
- **Providing Food**: This project also enables the robot to provide food for the pet

1.3 Challenges

- The IR sensor used won't pick up object over a certain distance. The distance of the sensor used here is very limited.

- The sensor used will pick up signals from every object in front of it not just the from the pet
- Failed to integrate video streaming to the blynk app.

1.4 Societal / Industrial Relevance

In our busy day to day life we don't have enough time to look after our pets.At that time a pet feeder robot becomes very handy.We can monitor our pet's action while we are at other places and also we can feed our pet's in the absence of their owner.

Chapter 2

Literature Survey

[1] "IoT Dog Daycare Robot with Dog Food Feeding"

The Robot is an IoT Based robot that is capable of taking care you your pets alone at home. The robot is integrated with a camera that allows for live streaming over IoT platform to get on demand footage of home. The robot takes monitoring far beyond a security camera as you can control the robot online over internet and move through your house any time you want.The robot dispenses appropriate amount of food in feeding tray as instructed by user online and then slides open the feeding tray, also it calls out your dog/cat by name to inform about feeding time. Once the pet has eaten it closes the feeding tray. All of this can be monitored online by the pet owner.

[2] "Human Following Robot using Arduino"

The human following robot using arduino project utilizes Arduino Uno, an ultrasonic sensor, and an L298N motor driver for obstacle detection and navigation. The robot, driven by code, halts and turns when obstacles are sensed, showcasing a basic form of autonomous behavior. This project encompasses hardware integration, software programming, and iterative testing to optimize performance. The goal is to create a versatile robot capable of dynamically navigating its environment

[3] "Automatic pet feeder using IoT"

A software code is dumped into the micro controller to perform operations like rotating motors and switching relays on and off. The whole feeder system is controlled using a mobile phone installed with blynk software. The user sends signals to the micro controller using blynk application through blynk cloud. When the DC servo motor runs, the motor rotates the propeller which is in the feeding device, which drops down pet food through the pipe into perforated feeding bowl

[4] “Live streaming Webcam using ESP32-CAM”

This repository showcases the development of a live video streaming web server using the ESP32-CAM module. The project demonstrates the successful implementation of a system that enables real-time video streaming from the ESP32-CAM to a web browser. The focus of this project was to leverage the capabilities of the ESP32-CAM module to create a web-based live video streaming application. By utilizing the ESP32-CAM's integrated camera module,so developed a system that captures video frames and streams them in real-time to a web browser. The project involved configuring the ESP32-CAM, establishing a local web server, and implementing the necessary functionalities for video capturing, encoding, and streaming. Through this implementation, users can access a web interface and view the live video stream from the ESP32-CAM.

Chapter 3

METHODOLOGY

3.1 Problem Statement

Many people face the problem of leaving their pets alone at the home while they leave for work or stay away from home for a period of time. This creates problem for the owners as it is difficult to feed the pet when left alone and can't monitor their pet all the time. We have created this rover with the aim of resolving this problem by providing a way to monitor their pets and to feed them effectively via the rover.

3.2 Block Diagram

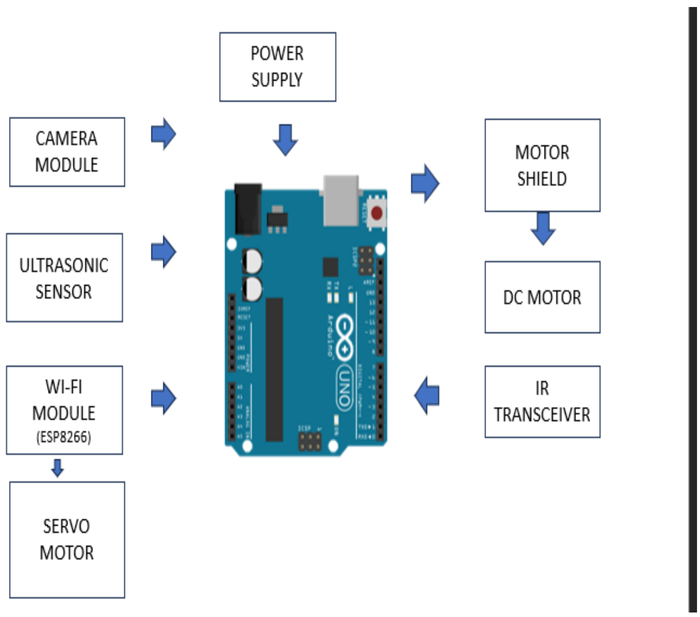


Figure 3.1: ESP32 CAM.

3.3 Working

3.3.1 Interfacing ESP 32 CAM with Arduino IDE

Step 1: Install Arduino IDE as well as the ESP 32 add-on.

Step 2:Insert the live streaming code to your IDE

Step 3:Make sure we have selected the Board "AI Thinker ESP32-CAM",GPIO 0 must be connected to GND to upload a sketch

Step 4:Before uploading the code, you need to insert your network credentials in the variables 'ssid' and 'password'

Step 5:After connecting GPIO 0 to GND, press the ESP32-CAM on-board RESET button to put your board in flashing mode.

Step 6: After uploading the code, disconnect GPIO 0 from GND. Open the Serial Monitor at a baud rate of 115200. Press the ESP32-CAM on-board Reset button for getting IP Address.

Step 7:Now, you can access your camera streaming server.

3.3.2 Interfacing ESP 8266 with servo motor

Step 1: Install Arduino IDE as well as the ESP 8266 add-on.

Step 2:Download the Blynk app and create a new template and obtain the Template ID.

Step 3: Create a new datastream in the app specifying the pins to control and angle for the servo to turn.

Step 4: Get the slider ready in the homepage for controlling the servo motor. Get the Auth code from the app.

Step 5:Implement the code to connect ESP 8266 and Blynk app by specifying the Template ID and Auth code in the code to connect both.

Step 6: Before uploading the code, you need to insert your network credentials in the variables 'ssid' and 'password'

. Step 7: Upload the code to ESP 8266 and after the ESP 8266 comes online, You can control it using the app.

3.3.3 Interfacing Arduino Board

Step 1: Implement the Arduino Uno using the Arduino IDE Compiler.

Step 2: Connect the Ultrasonic sensor and IR sensor to the Arduino Uno which is connected to the motor driver which drives the motor of the rover for the rover to move.

Step 3: Upload the code for Object Following Rover to the Arduino Uno.

3.4 Implementation

Once the device is turned on, the IR sensor detects the pet and transmit this signal to Arduino Uno which analyze it and gives signal to motor driver to drive the motors. The ultrasonic sensor helps to maintain distance and not to ram on to the pet. The ESP 8266 is connected to the Blynk app where we control the servo motor which helps to provide food for the pet. The ESP 32 camera always keep streaming video as long as power and network is provided to it.

3.5 Circuit Diagrams

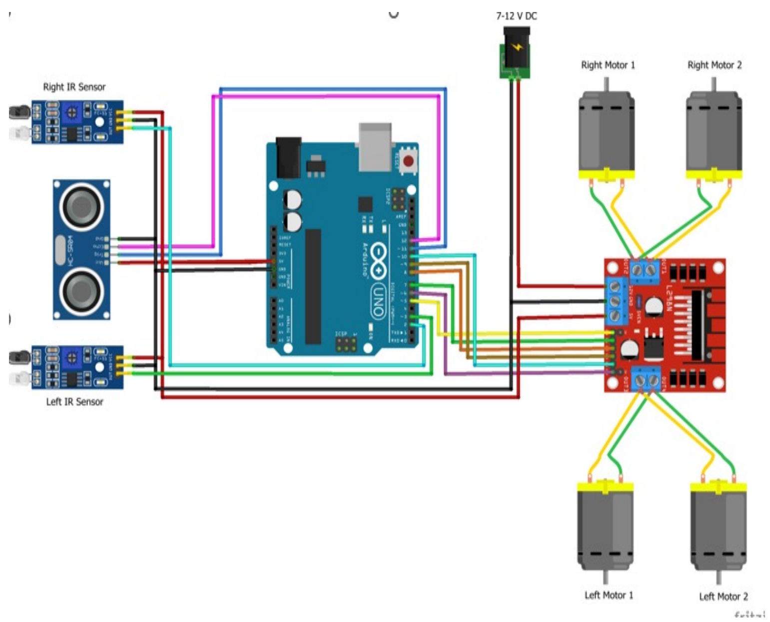


Figure 3.2: Object Flowing Robot using Arduino

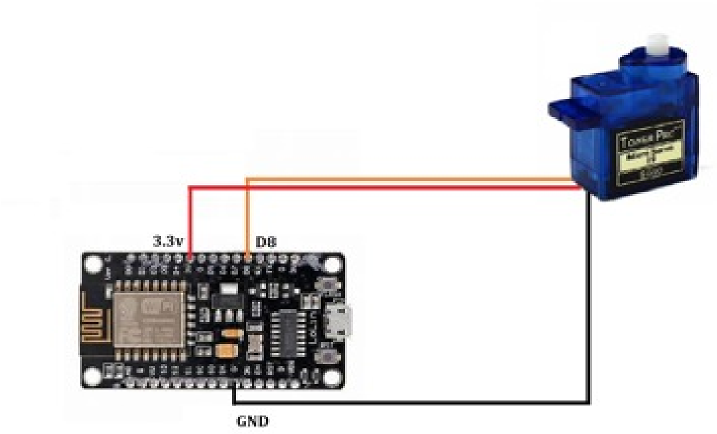


Figure 3.3: ESP8266 WiFi Module with Servo Motor

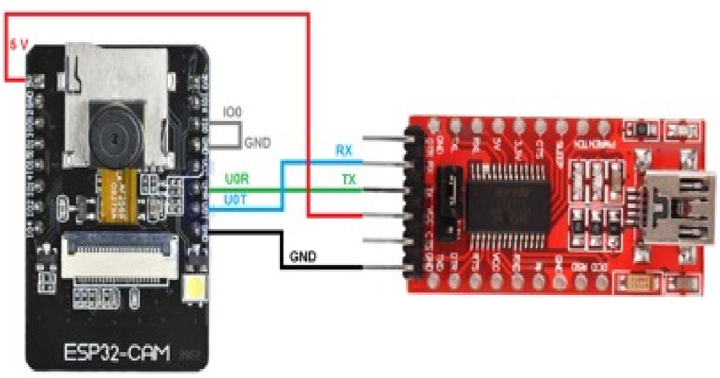


Figure 3.4: ESP32 CAM using FTDI Programmer

3.6 Flowchart

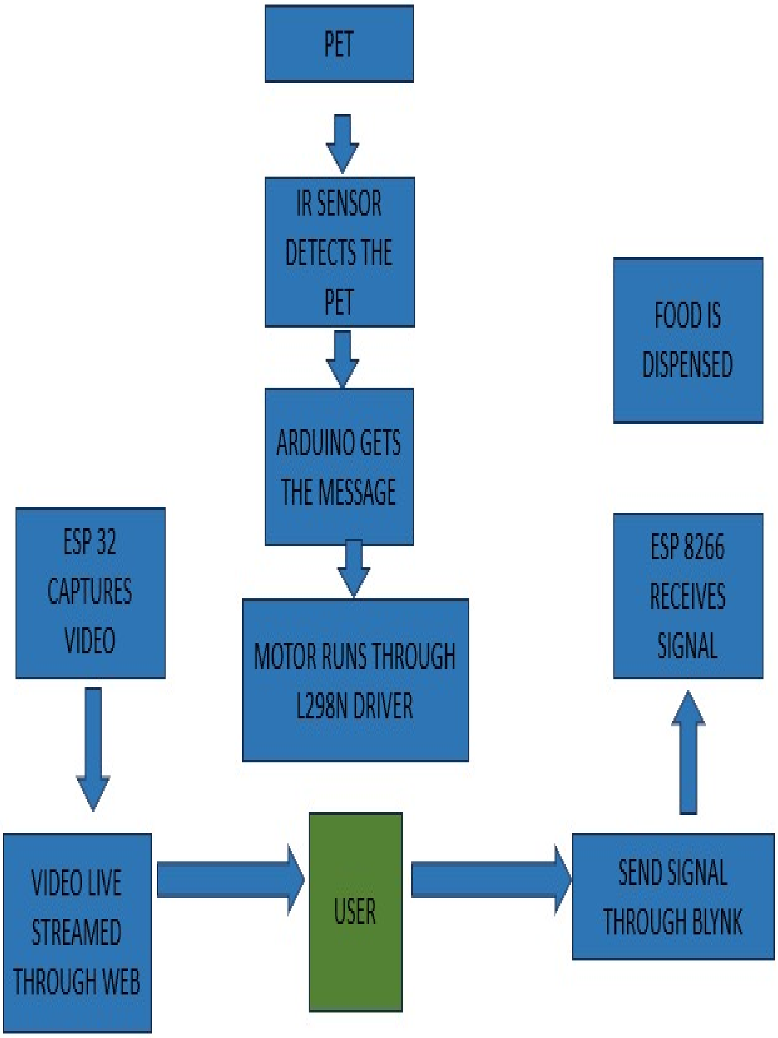


Figure 3.5: Flowchart

Chapter 4

HARDWARE AND SOFTWARE DETAILS

4.1 Hardware Details

4.1.1 Arduino Uno

A microcontroller board based on the ATmega328P. It has 14 digital input/output pins, 6 analog inputs, and a 16 MHz quartz crystal. Acts as the main controller to process inputs from sensors and control outputs like motors and servos.It can run on both online and offline platforms.It works at 5V with 14 digital I/O pins,6 of which are PWM o/p,while the others are to provide accurate data from sensors. Arduino Uno has a clock speed of 16 MHz, which makes it suitable for real-time processing tasks.It offers 32 KB flash memory for programming codes storage,SRAM –2 KB and EEPROM – 1 KB where information can be stored when power goes off. In terms of the power source, the unit can work with any voltage ranging between 6 V and 20 V or even within a range of from about 7V to around12V The Arduino Uno features various communication interfaces like UART, SPI, and I2C enabling it to communicate with other peripherals as well as other microcontrollers.Additionally,there is an embedded USB interface also based on ATmega16U2 controller to aid in easier programming and transfering of data.

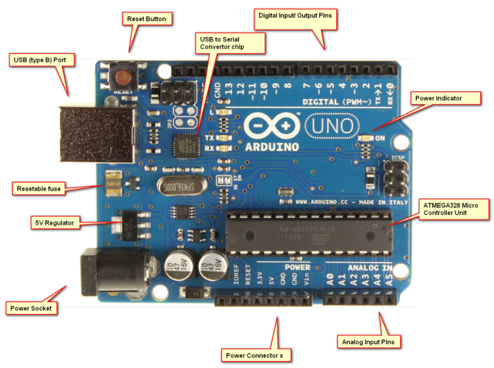


Figure 4.1: Arduino UNO

4.1.2 Ultrasonic Sensor HC-SR04

HC-SR04 Ultrasonic (US) sensor is a 4 pin module,whose pin names are Vcc, Trigger,Echo and Ground respectively. This sensor is a very popular sensor used in many applications where measuring distance or sensing objects are required.The module has two eyes like projects in the front which forms the Ultrasonic transmitter and Receiver.The Ultrasonic transmitter transmits an ultrasonic wave, this wave travels in air and when it gets objected by any material it gets reflected back toward the sensor this reflected wave is observed by the Ultrasonic receiver module.

Figure 4.2: Ultrasonic Sensor HC-SR04

4.1.3 IR Sensor

IR sensor is an electronic device, that emits the light in order to sense some object of the surroundings.An IR sensor can measure the heat of an object as well as detects the motion.Usually, in the infrared spectrum, all the objects radiate some form of thermal radiation.The emitter is simply an IR LED (Light Emitting Diode) and the detector is simply an IR photodiode. Photodiode is sensitive to IR light of the same wavelength which is emitted by the IR LED.When IR light falls on the photodiode,the resistances and the output voltages will change in proportion to the magnitude of the IR light received.

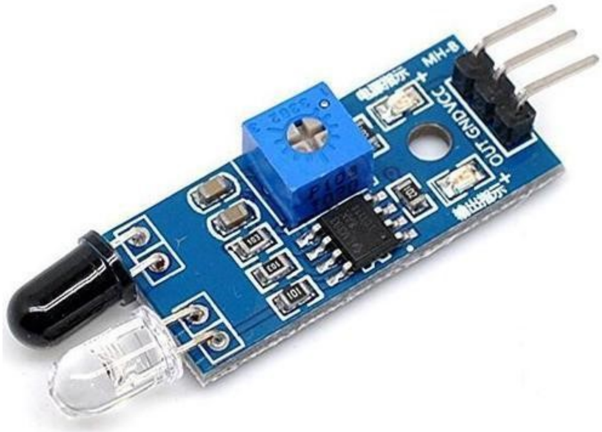


Figure 4.3: IR Sensor.

4.1.4 Servo Motor SG90

It is a type of rotary actuator that provides precise control over the angular position of a device.The SG90 is a 9g micro servo motor with a torque of 1.8 kg/cm and a rotation angle of approximately 180 degrees. The main application of SG90 Servo Motors is in small-scale hobby projects such as remote-controlled cars, robots, and aircraft.

Figure 4.4: Servo Motor.

4.1.5 L298N Motor Driver

L298N module is a high voltage, high current dual full-bridge motor driver module for controlling DC motor and stepper motor.It can control both the speed and rotation direction of two DC motors.This module consists of an L298 dual-channel H-Bridge motor driver IC.This module uses two techniques for the control speed and rotation direction of the DC motors.

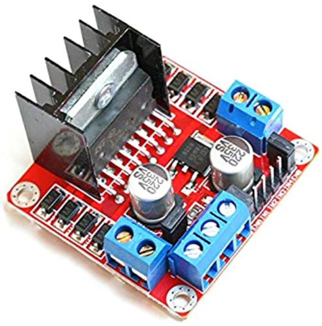


Figure 4.5: L298N Motor Driver.

4.1.6 ESP32-CAM

The ESP32-CAM is a full-featured microcontroller that also has an integrated video camera and microSD card socket.It's inexpensive and easy to use, and is perfect for IoT devices requiring a camera with advanced functions like image tracking and recognition.This board has 4MB PSRAM which is used for buffering images from the camera into video streaming or other tasks and allows you to use higher quality in your pictures without crashing the ESP32.It also comes with an onboard LED for flash and several GPIOs to connect peripherals.

Figure 4.6: ESP32 CAM.

4.1.7 FTDI Programmer

The USB to RS232 converter cables provide a simple communication method between serial devices with RS232 to modern USB supported devices.FTDI cable comes with an internal mounted electronic circuit which uses FTDI FT232R chip. This FTDI chip converts USB data to serial and vice versa.In other words,this cable provides an effective and cheap solution to connect TTL serial interface to USB.To use this FTDI cable,we need a device driver which is available to download freely from FTDI website.After installation of device drivers,the US232R adapter appears as a virtual COM port in device manager settings. [!h]

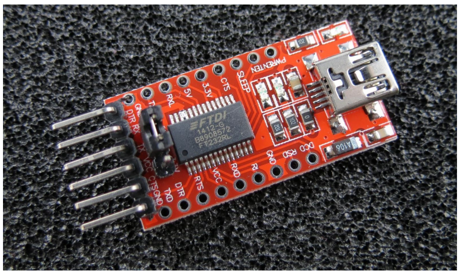


Figure 4.7: FTDI Programmer.

4.1.8 ESP8266 WIFI Module

It is used to enable the internet connection to various applications of embedded systems. Espressif systems designed the ESP8266 Wi-Fi module to support both the TCP/IP capability and the microcontroller access to any WI-FI network.The ESP8266 Wi-Fi module is highly integrated with RF balun, power modules,RF transmitter and receiver, analog transmitter and receiver,amplifiers, filters,digital baseband, power modules, external circuitry, and other necessary components.The ESP8266 Wi-Fi module comes with a boot ROM of 64 KB, user data RAM of 80 KB, and instruction RAM of 32 KB. It can support 802.11 b/g/n Wi-Fi network at 2.4 GHz along with the features of I2C, SPI, I2C interfacing with DMA, and 10-bit ADC.

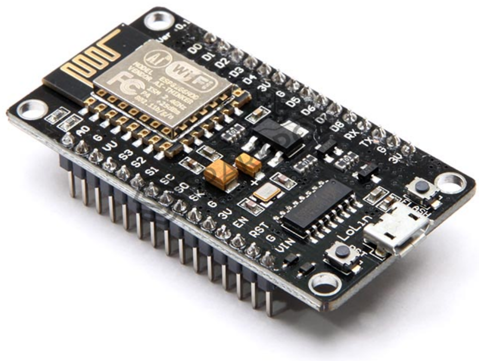


Figure 4.8: ESP8266 WIFI MODULE.

4.2 Software Details

4.2.1 Arduino Compiler

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code,a message area,a text console, a toolbar with buttons for common functions and a series of menus.It connects to the Arduino hardware to upload programs and communicate with them.The IDE includes a complete library a it contains built-in functions and example sketches to help accelerate project development and easy integration with sensors, actuators and modules. The IDE makes it easy to compile code directly and run it via a USB connection to the Arduino board. Its cross-platform compatibility (Windows, macOS and Linux) and strong community support make the Arduino IDE an essential tool for a wide range of electronics and IoT projects.

Figure 4.9: Arduino Compiler.

4.2.2 Blynk IoT

Blynk is a platform for the development of smartphone applications that work with a wide range of microcontrollers.The main focus of the Blynk platform is to make it super-easy to develop the mobile phone application. As you will see in this course, developing a mobile app that can talk to your Arduino is as easy as dragging a widget and configuring a pin.

Figure 4.10: BLYNK IoT.

Chapter 5

Results

This project has succesfully developed a method to efficiently monitor the pets daily by following them everywhere they go and dispense right amount of food as required by the user.

Figure below shows the final phase of the rover.

5.1 Camera output

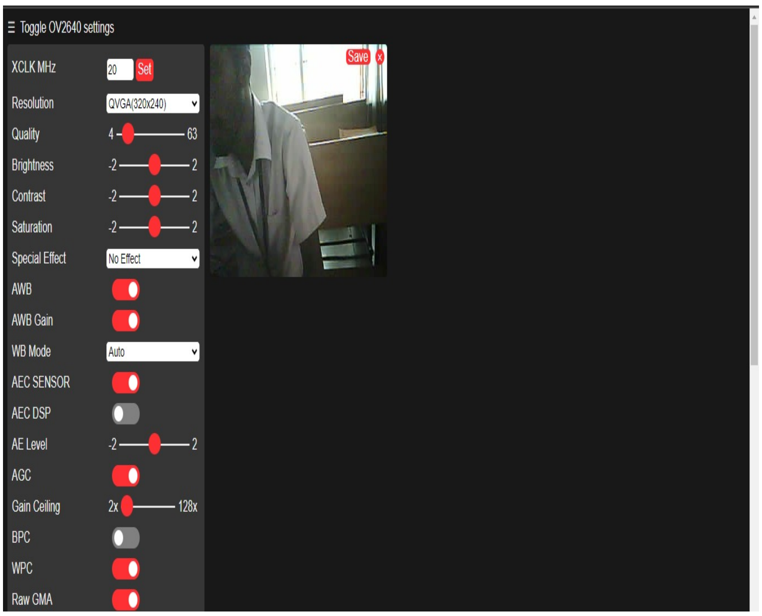
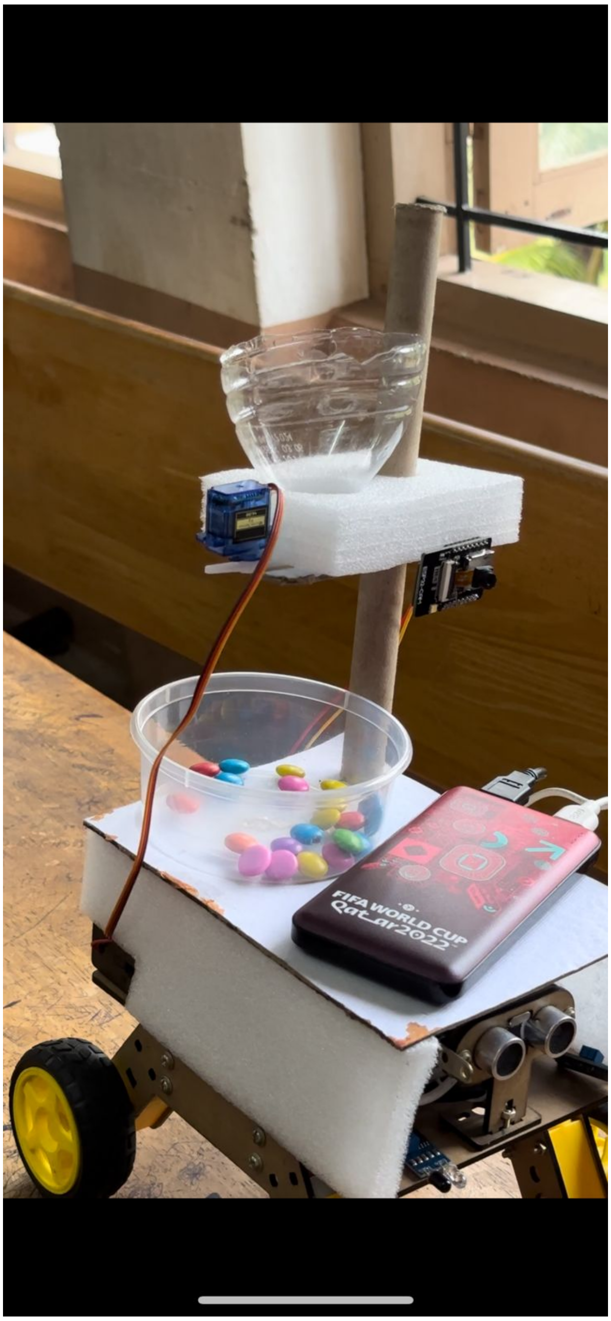
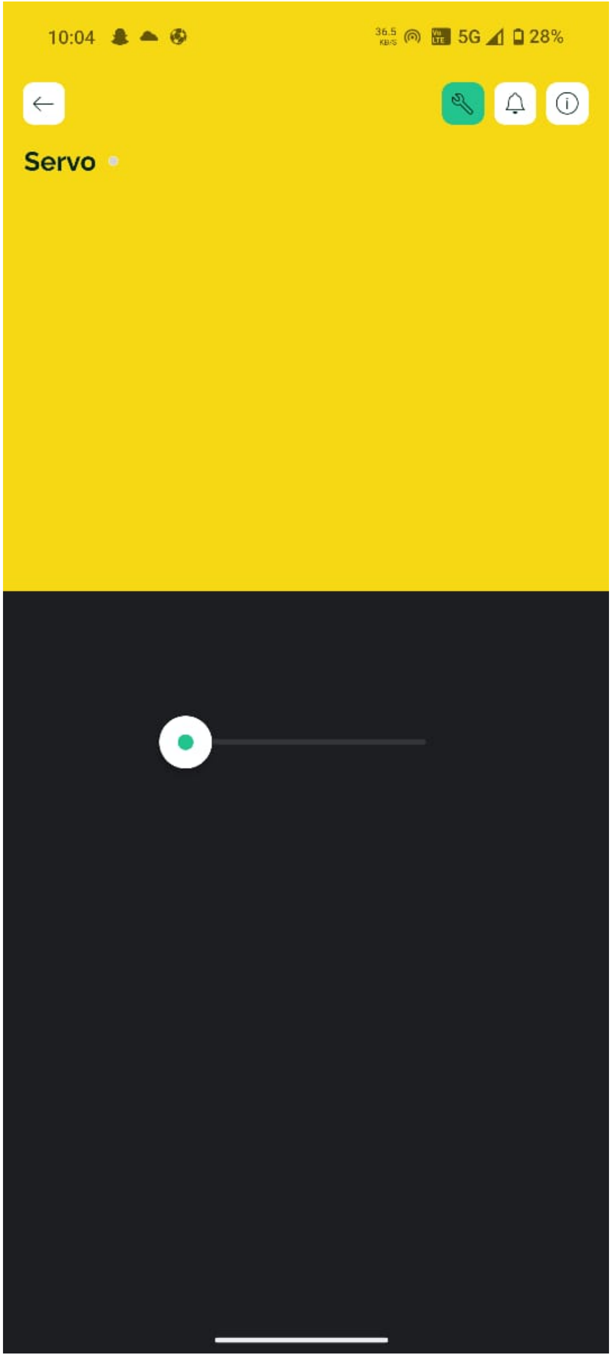


Figure 5.1: Camera Output from the Webserver

5.2 Food dispenser

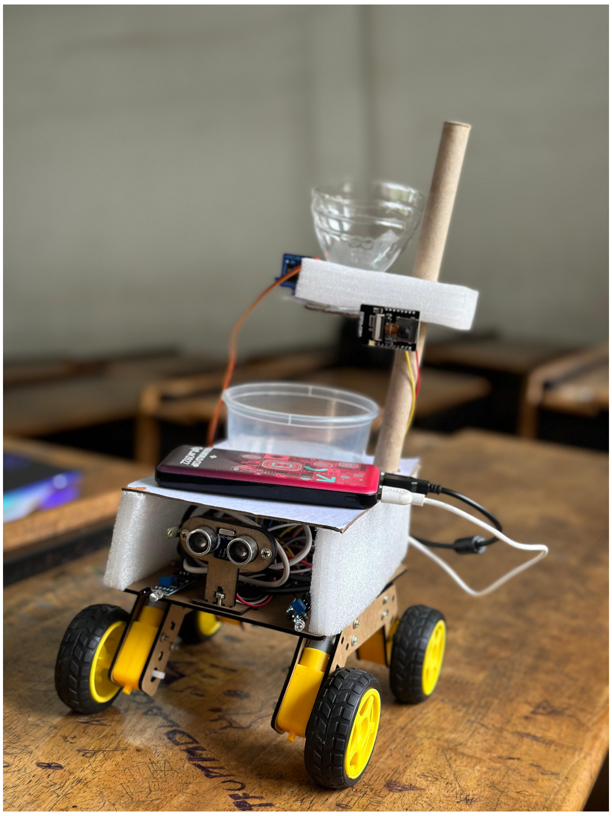


(a) Food falling down



(b) Slider movement for Blynk App

5.3 Final Output



Chapter 6

Conclusions & Future Scope

6.1 Conclusion

The robot dispenses appropriate amount of food in feeding tray as instructed by user online and so open the funnel to dispense the food. All of this could be monitored online by the pet owner. This IoT based pet feeder are able to feed pet efficiently and also save the animals or pets from starvation and can follow them so we can have constant monitoring.

6.2 Future Scope

The implementation of two motors to dispense a ball a few meters can help the pet to play fetch and keep them entertained. The addition of speaker can help us interact with the pet through voice in real time. A lighter structure frame can help us reduce the weight and can help the rover move more freely. The use of Bluetooth can reduce many problems we are facing right now as they are more effective in following the pet. Better IR sensors can help us significantly in navigating through the terrain.

Chapter 7

Mapping CO and Project Objectives with PO-PSO

The mapping for different program outcomes, program educational objectives and program specific outcomes to the project work outcomes are shown in this chapter. **The course outcomes are as the following:**

CO 1: Be able to practice acquired knowledge within the selected area of technology for project development.

CO 2: Identify, discuss and justify the technical aspects and design aspects of the project with a systematic approach.

CO 3: Reproduce, improve and refine technical aspects for engineering projects.

CO 4: Work as a team in development of technical projects.

CO 5: Communicate and report effectively project related activities and findings

Mapping : The following tables shows the mapping for course outcome for PO,PEO and PSO.The mapping specification are as follows,

1-Low,2-Medium,3-High

CO	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**
1	3	3	3	2	-	3
2	3	3	3	2	-	3
3	3	3	3	2	-	3
4	-	-	-	-	-	-
5	-	-	-	-	-	-

Table 7.1:Mapping of course outcomes for PO1-PO6.

CO	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
1	3	3	3	2	-	3
2	3	3	3	2	-	3
3	3	3	3	2	-	3
4	-	-	-	-	-	-
5	-	-	-	-	-	-

Table 7.2:Mapping of course outcomes for PO7-PO12.

CO	PSO 1	PSO 2	PSO 3
1	2	1	-
2	2	2	-
3	2	2	-
4	2	1	1
5	2	1	1

Table 7.3:Mapping of course outcomes for PSO1-PSO3

Mapping and justification of project objectives /outcomes with PO and PSOs:The table 4 shows mapping of project objectives with POs and PSOs.Tables 5 and 6 shows the justification of project outcome mappings with POs and PSOs respectively.

CO	PROJECT OBJECTIVES	POs	PSOs
1	System provides real time continuous monitoring of the pet	3	3
2	It retraces the path taken by the pet and follows it	3	2
3	System provides the required amount of food declared by the user	3	3
4	The project was set to be under a strict budget	2	1
5	Optimized Design and user comfort.	2	2

Table 7.4:Mapping of Project Objectives with POs and PSOs

CO - PO Mapping

Project Outcomes	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	Justification
Learned how to use Blynk software	2	3	1	1	3	1	2	Acquired skills in Blynk App
Learnt how to use ESP-32 and ESP 8266	3	3	3	3	2	2	2	Acquired skills in ARDUINO UNO IDE coding and interfaced with ESP-32
Hardware implementation	3	2	3	2	1	-	-	Acquired skills to work in tech-related groups.
Documentation and presentation	-	-	-	-	2	-	-	Preparing the project report and presenting our findings helped us enhance our presentation and communication skills

Table 7.5: Justification for mapping Project outcomes with POs

CO - PO Mapping

Project Outcomes	PO 8	PO 9	PO 10	PO 11	PO 12	Justification
Learned how to use Latex software	2	3	3	3	3	Acquired skills in Latex
Learnt how to use ESP-32	2	3	2	3	3	Acquired skills in ARDUINO UNO IDE coding and interfaced with ESP-32
Hardware implementation	-	3	2	2	3	Acquired skills to work in tech-related groups
Documentation and presentation	-	-	3	-	2	Preparing the project report and presenting our findings helped us enhance our presentation and communication skills

Table 7.6: Justification for mapping Project outcomes with POs

Project Outcomes	PSO 1	PSO 2	PSO 3	Justification
Learned how to use Blynk software	2	3	3	Acquired skills in Blynk App
Learnt how to use ESP-32 and ESP-8266	3	2	3	Acquired skills in ARDUINO UNO IDE coding and interfaced with ESP32 and ESP 8266
Hardware implementation	3	3	3	Acquired skills to work in technical environment
Documentation and presentation	-	2	3	Preparing the project report and presenting our findings helped us enhance our presentation and communication skills

Table 7.7: Justification of mapping of Project Outcomes with PSOs

Appendix A: Questions and Answers

Qn1) How does the rover follow the pet?

This project is uses IR Proximity Sensors and Ultrasonic Sensor to locate and track an object within its proximity. This signal is sent to motor driver through Arduino Uno which tells the motor driver on which motor to run and when and this is how the rover follows the pet.

Qn2) What is the range for the Wi-Fi module?

The module has a range of about 50 meters. But the range doesn't matter since the it travels with the rover. It's functional as long as power is given to the module.

Qn3) Why didn't you combine both food dispense control and monitoring system into a single software?

We first decided to do the same by integrating both into the Blynk App where the user can control the servo motor and stream the live video within the same app. But Blynk has recently decided to make the live streaming function a premium feature which require us to pay the premium to use that function.

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